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CRYPTO

5 LESSONS

PHASE 4 · CLOUD, DEVSECOPS & CRYPTANALYSIS

Cryptographic Attacks & Cryptanalysis

Breaking weak crypto where it's misused: brute force, rainbow tables, timing attacks, padding oracle, hash collisions, side channels, and quantum-era threats.

01

Cryptographic Misuse Patterns

INTERMEDIATE

CONCEPT

40 min →

Most crypto 'breaks' come from misuse not math: weak ciphers, ECB, sh...

02

Brute Force, Rainbow Tables & Password Hashes

INTERMEDIATE

LAB

45 min →

Search space math, GPU cracking economics, salted-hashing, rainbow ch...

03

Timing Attacks & Side Channels

ADVANCED

LAB

45 min →

Timing differences leak secrets: login user-enum, toString overhead, ...

04

Hash Collisions, Length Extension & Block-chain Games

ADVANCED

LAB

45 min →

Collision attacks, birthday bounds, preimage ambitions, and the lengt...

05

Quantum Threats & Post-Quantum Cryptography

ADVANCED

CONCEPT

35 min →

Shor vs RSA/ECC, Grover vs symmetric key, harvest-now-decrypt-later -...

